Performance Comparison of Proof of Authority and Proof of Stake Consensus Mechanisms on DChain National Blockchain Network

1st Maulana Anjari Anggorokasih
Department of Electrical and Information Engineering
Faculty of Engineering
Universitas Gadjah Mada
Yogyakarta, Indonesia
maulana.anjari2903@mail.ugm.ac.id

2nd Noor Akhmad Setiawan
Department of Electrical and Information Engineering
Faculty of Engineering
Universitas Gadjah Mada
Yogyakarta, Indonesia
noorwewe@ugm.ac.id

3rd Teguh Bharata Adji
Department of Electrical and Information Engineering
Faculty of Engineering
Universitas Gadjah Mada
Yogyakarta, Indonesia
adji@ugm.ac.id

***Abstract*—DChain is Indonesia's national blockchain infrastructure for higher education electronic certificates, currently operating on Proof of Authority (PoA) with planned migration to Proof of Stake (PoS) to enhance decentralization. This research aims to evaluate and compare the performance of both consensus mechanisms to determine migration feasibility. We employed a quantitative experimental approach through computerized benchmarking using Hyperledger Caliper in a distributed multi-node environment (Docker and Kubernetes). Testing was conducted across various scenarios including throughput, latency, computational resource efficiency, and network stability (soak test). Results indicate that Proof of Stake possesses higher throughput capacity (≈60 TPS) compared to PoA (≈30–40 TPS). However, PoA proved superior in responsiveness, with average latency of 4.6 seconds compared to 8 seconds for PoS, and demonstrated better CPU utilization efficiency. We conclude that the PoA consensus mechanism remains the most optimal solution for DChain's current requirements due to the user-interactive nature of academic services that necessitate rapid response, while PoS offers scalability potential for long-term development.**

***Index Terms*—Blockchain, DChain, Proof of Authority, Proof of Stake, Benchmarking**

I. Introduction

Blockchain technology, first introduced through Bitcoin in 2009, has evolved significantly beyond cryptocurrency applications. The introduction of Ethereum in 2015 with smart contract capabilities expanded blockchain's potential into various sectors. This evolution has driven extensive research into alternative consensus mechanisms more efficient than Proof of Work (PoW), particularly Proof of Authority (PoA) and Proof of Stake (PoS).

In Indonesia, DChain (Dikti Chain) represents a national blockchain network designed to support the higher education ecosystem. DChain functions as a decentralized digital infrastructure enabling the issuance, storage, and verification of academic documents—such as digital diplomas, competency certificates, and transcripts—with guaranteed data integrity. A crucial component determining DChain's performance is the consensus mechanism used to validate transactions and maintain network integrity.

Currently, DChain implements PoA consensus, known for its efficiency as only authorized nodes can validate transactions. However, increasing scale and network complexity necessitate a more decentralized and scalable mechanism like PoS, which allows all token holders to participate in the validation process. This issue aligns with global trends, where Ethereum, one of the largest public blockchains, migrated from PoW to PoS in 2022.

The planned migration from PoA to PoS in DChain raises strategic questions regarding its impact on network performance. To date, no comprehensive study has evaluated consensus migration on DChain using standardized testing parameters. Therefore, this research is designed to analyze and compare DChain's performance under both consensus mechanisms, focusing on metrics such as Transactions Per Second (TPS), latency, computational resource usage, scalability, and system stability.

This research is important to fill this gap by providing empirical foundations for policymakers and developers in determining optimal consensus strategies to support digital transformation in Indonesian higher education.

II. Related Work

Several studies have examined the performance of PoA and PoS in blockchain contexts. Choi and Hong [1], Rebello et al. [2], and Arslan et al. [3] compared both consensus mechanisms based on basic metrics such as Transactions Per Second (TPS) and latency. Their research showed that PoA has advantages in low latency and high throughput under moderate loads, while PoS excels in decentralization and performance at large transaction scales. However, testing was limited to basic performance parameters without analysis of resource usage and system stability.

From a methodological perspective, Li [4], Kumar et al. [5], and Kaushal & Kumar [6] emphasized the importance of using

benchmarking tools like Hyperledger Caliper for blockchain performance evaluation. This tool enables measurement of standard metrics such as TPS, latency, and resource consumption. Nevertheless, most research only tested one consensus mechanism at a time, thus not providing a comprehensive direct comparison between PoA and PoS.

Scalability and resource efficiency aspects have also been addressed in previous research. Sharma et al. [7] and Ali et al. [8] found that PoA tends to be more CPU and memory efficient for low to medium loads, while PoS is more stable when facing transaction load variations. However, these studies were generally based on simulations on private Ethereum networks and were not related to the education sector context.

In the educational context, Gaikwad & Patel [9] and Singh & Chatterjee [10] emphasized blockchain's role in digital diploma verification and academic data management. Implementations in these studies predominantly used PoA due to its efficiency, but did not examine the potential use of PoS more deeply.

The research most closely approaching this study's focus is by Wicaksono, Hatta, and Aristyagama (2025) [11]. They compared PoA (Ethereum 1.0) and PoS (Ethereum 2.0) on blockchain-based academic transcript systems. Testing was conducted with Hyperledger Caliper using throughput and latency metrics. Results showed that PoA was superior under low to medium loads, while PoS was more suitable for large-scale transactions. However, this research had important limitations: (1) only using two metrics (TPS and latency) without covering resource usage; (2) experiments conducted with one worker in Caliper thus not reflecting multi-user conditions; (3) focus limited to one application (academic transcripts); and (4) no system stability analysis including fault tolerance and fork rate.

This research fills these gaps by expanding testing scope to include resource usage metrics, scalability, and system stability. Additionally, this research applies more diverse benchmarking parameter variations, including number of nodes, block time, block size, transaction load level, transaction operation types, and number of workers. Research focus is directed at DChain as a national blockchain infrastructure for higher education.

III. Proposed System and Methodology

A. Infrastructure and Software Stack

The research infrastructure consists of:

- 1× Control Plane VPS (AMD EPYC, 8 vCPU, 24GB RAM, 200GB NVMe) hosting Caliper orchestrator, Prometheus, Grafana, and PostgreSQL
- 5× Worker VPS (heterogeneous: 2× AMD EPYC, 3× Intel Broadwell, 6 vCPU, 12GB RAM, 200GB SSD each) functioning as worker nodes running PoA and PoS consensus protocols

Software stack includes:

- PoA: Go-Ethereum v1.13.15 + Clique consensus
- PoS: Go-Ethereum v1.16.7 + Prysm v6.0.4 (Gasper consensus: LMD-GHOST + Casper FFG)
- Benchmarking: Hyperledger Caliper v0.6.0
- Monitoring: Prometheus + Grafana + Node Exporter
- Orchestration: Kubernetes (k3s) + Kurtosis (for PoS deployment)

The technical implementation uses a combination of existing open-source software (given) and components developed by the researcher. Given software includes Geth, Prysm, Kurtosis, Hyperledger Caliper, and Kubernetes. Self-developed components include bash and Docker-based PoA network generation scripts, Starlark configuration for PoS network orchestration, automated benchmarking pipeline (`run_pipeline.sh`), and centralized monitoring system configuration with Dual-Mode Scraping mechanism. All source code is publicly available at the repository listed in the Acknowledgment section.

B. Network Topology

The experimental topology employs a 10-node configuration distributed across 6 VPS instances, as illustrated in Figure 1. The infrastructure consists of:

- **Control Plane VPS**: Orchestration center running Kubernetes Master, Docker, Prometheus (metrics collector), Grafana (visualization), Ethstats (Ethereum network monitoring), Kurtosis (PoS orchestration), PostgreSQL (data storage), and Hyperledger Caliper (transaction load generator)
- **Validator Nodes**: 5 nodes (black border) with rights to validate transactions and produce new blocks. In PoA consensus, these nodes use Round-Robin proposer rotation; in PoS, validator selection is randomized
- **Observer Nodes**: 5 nodes (red border) that participate in data synchronization without validation rights, functioning as RPC gateways for transactions sent by Caliper

Each worker VPS runs Kubernetes Worker, Pushgateway, Node Exporter, and cAdvisor to collect performance metrics forwarded to Prometheus on the Control Plane. The orange dashed arrows indicate RPC Load flow from Hyperledger Caliper to Observer Nodes for benchmarking tests.

C. Benchmarking Scenarios

Six main scenarios were applied to test various performance aspects following the 6-Layer Benchmarking Strategy:

1) **Baseline Performance**: Measure intrinsic system latency under minimal load to establish baseline metrics
2) **Stress Testing (Step Load)**: Find system saturation point (knee point) by gradually increasing TPS load (5, 10, ..., 100 TPS) until latency spikes drastically
3) **Scalability Testing**: Test concurrent client handling efficiency by running optimal TPS load with variations of 1, 3, and 5 workers, and measure read operation (RPC) performance with high read loads (200, 400, 600 TPS)
4) **Resource Efficiency**: Measure computational resource usage (CPU, RAM, Disk I/O, Network) to evaluate infrastructure efficiency

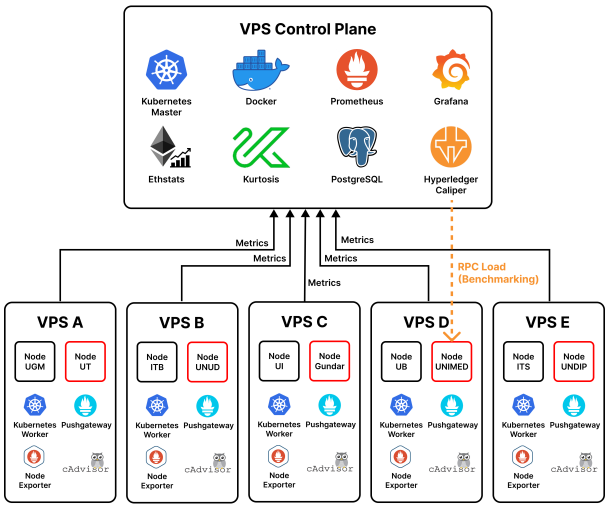

Fig. 1. Testbed Infrastructure Architecture: Distribution of Software Components Across VPS Instances

5) **Resilience & Stability (Soak Test)**: Test long-term stability through Soak Test for 15 minutes at 60% optimal load to detect cumulative performance degradation
6) **Functional Lifecycle**: Validate data consistency through complete transaction lifecycle (Mint-Read-Burn)

D. Metrics

Performance metrics measured include:

- **Transaction Performance**: Throughput (TPS), Latency (avg, P95), Success Rate
- **Resources**: CPU Usage (%), Memory Usage (MB), Network I/O, Disk I/O
- **Stability**: Stability Index (TPS consistency in long duration)

E. Monitoring and Data Collection

The monitoring system was built using Prometheus collecting metrics from three sources:

1) **Geth/Prysm**: Internal blockchain metrics (peer count, block height, tx pool)
2) **Node Exporter/cAdvisor**: Infrastructure metrics (CPU, RAM, Network container)
3) **Hyperledger Caliper**: Transaction metrics (TPS, Latency, Errors) sent via Pushgateway

Data was stored in PostgreSQL for further analysis.

The high-level architecture of Hyperledger Caliper used in this research is illustrated in Figure 2. The architecture shows the workflow from configuration inputs (Workload Module, Benchmark Configuration, Network Configuration) through the Caliper engine to the System Under Test (SUT), producing benchmark reports as output [12].

IV. Results and Discussion

A. Throughput and Saturation Point

Throughput Step testing increased load from 5 to 100 TPS. Results showed that in 10-node topology, PoA began

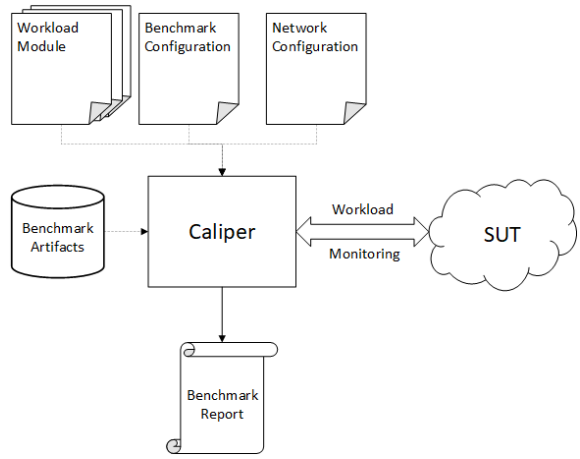

Fig. 2. High-level Architecture of Hyperledger Caliper (Source: Official Documentation)

experiencing saturation at load ≈30–40 TPS, while PoS maintained linearity up to ≈60 TPS before saturation (see Figure 3). PoS achieved peak throughput higher than PoA in this configuration, with the comparison of optimal stability points summarized in Figure 4.

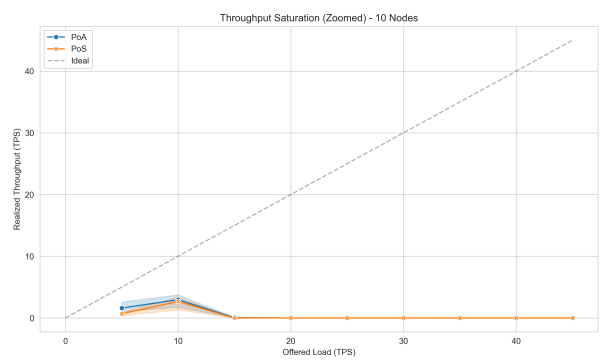

Fig. 3. Throughput Curve on 10-Node Topology

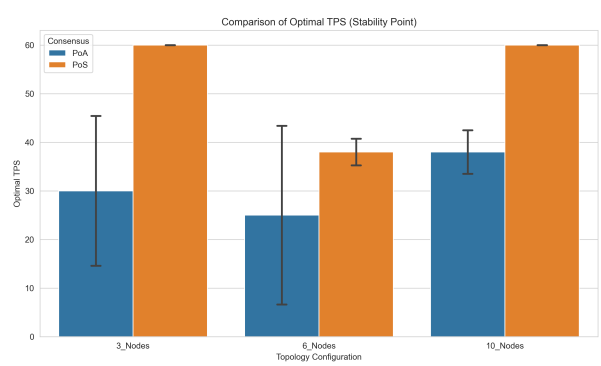

Fig. 4. Optimal TPS Comparison

B. Transaction Latency

Latency measurement was conducted under steady-state conditions. The latency trade-off graph (Figure 5) shows the relationship between load (TPS) and response time.

- **Baseline Latency**: At low load (< 15 TPS), PoA has average latency of 4–5 seconds, while PoS ranges 7–8 seconds
- **Latency Under Load**: When approaching saturation point, PoS latency increases sharply compared to PoA which tends to have more stable increases

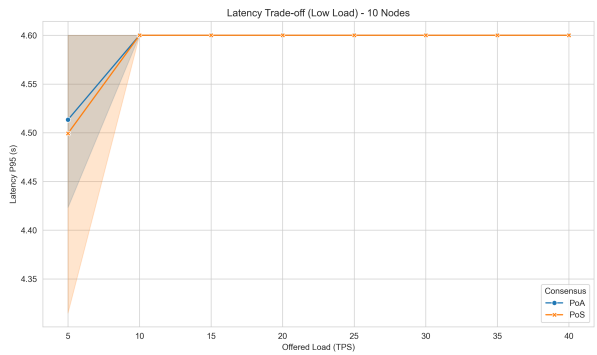

Fig. 5. Latency Trade-off (Hockey Stick Curve) on 10-Node Topology

C. Resource Efficiency and Scalability

Efficiency was measured based on TPS ratio per percent CPU usage, and scalability was measured by adding number of client workers. Results showed PoA has much higher CPU efficiency, as visualized in Figure 6. PoS has significant computational overhead. On client scalability side, both consensuses experienced performance degradation when number of workers increased from 1 to 5, indicating bottleneck on client/RPC side.

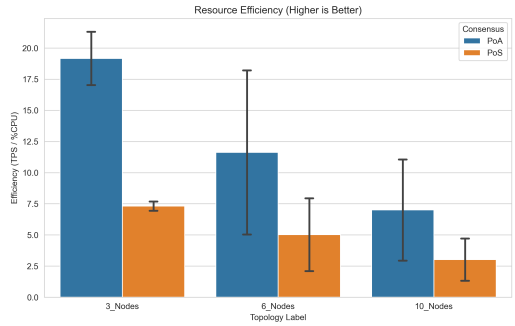

Fig. 6. Resource Efficiency (TPS per % CPU)

D. Analysis

Data shows an interesting phenomenon where PoS actually achieves higher saturation throughput than PoA at 10 Nodes scale. This is suspected to be caused by Block Time limitations. PoA with 5-second blocks has a physical limit on gas amount/block per time unit. Conversely, PoS with 12-second slots, although the interval is longer, allows more efficient transaction batching with the same gas limit per block, producing higher aggregate throughput before RPC saturation.

PoA shows clear superiority in responsiveness aspect. PoA average latency (4.6 seconds) is very close to block time (5 seconds), showing minimal inefficiency. Conversely, PoS (8 seconds) suffers from inherent latency from slot mechanism and initial probabilistic finality. For academic service systems, this 3–4 second difference is significant in terms of UX (User Experience).

PoA excels in resource efficiency. PoS "pays" for its decentralization with expensive computation for attestation. Both systems' stability proved robust in long-term testing (Soak Test), with no indication of memory leaks found.

The overall performance profile is summarized in Figure 7, showing the trade-off between the two mechanisms across five key dimensions.

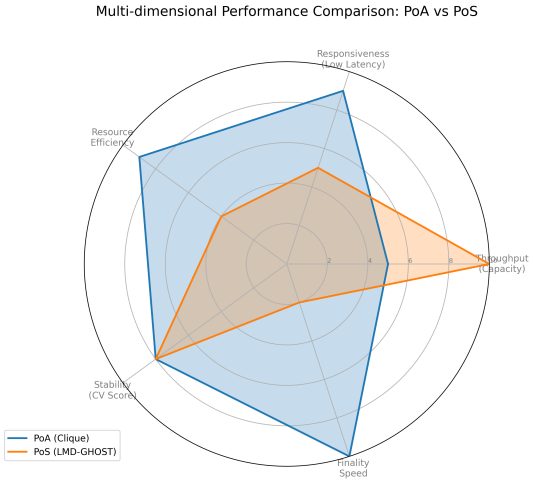

Fig. 7. Radar Chart Comparison: PoA vs PoS Multidimensional Trade-off

E. Synthesis: Relevance for DChain

For academic certificate use cases in DChain, **PoA** characteristics proved more advantageous because:

1) **User-Interactive**: Low latency provides better user experience
2) **Read-Heavy**: Write transaction volume (diploma issuance) is relatively low and seasonal, so PoS massive capacity is less utilized compared to PoA operational cost efficiency
3) **Finality**: PoA Clique offers deterministic finality that is simpler for application integration

F. Comparison with Previous Research

This research provides empirical contribution by comparing DChain testing results with existing literature.

This research fills the gap (research gap) with:

1) **Specific Case Study**: Testing functionally relevant workload (Lifecycle), not just token transfer

TABLE I
COMPARISON OF DCHAIN RESULTS WITH PREVIOUS LITERATURE

Aspect	DChain Findings	Prior Research	Analysis
Throughput	≈30–40 TPS (Early Saturation)	PoA > 1000 TPS [1], [2]	DChain results lower due to limited VPS resources and real geographic latency
Latency	≈4–5 seconds	PoA < 1 second [3], [11]	Consistent with 5-second block time configuration
Efficiency	Very High (TPS/%CPU)	Considered efficient [7], [8]	Validates finding that PoA is much lighter than PoS

2) **Real Geographic Topology**: Simulation of inter-campus latency (UGM, UI, ITB), not localhost
3) **Stability Analysis**: Adding Soak Test to prove system resilience

In summary, the comprehensive benchmarking results provide a clear technical justification for DChain's strategic consensus roadmap. By filling the research gaps with geographically distributed testing and functionally relevant workloads, this study establishes that while PoS offers superior raw throughput, PoA's latency and resource efficiency are more aligned with the immediate needs of academic certificate verification. This empirical foundation ensures that any future protocol migration within the national consortium can be executed with a precise understanding of the performance trade-offs, ultimately sustaining the quality of digital services in the Indonesian higher education ecosystem.

V. CONCLUSION

This research aimed to evaluate and compare the performance of Proof of Authority (PoA) and Proof of Stake (PoS) consensus mechanisms in the context of implementing higher education electronic certificate infrastructure (DChain). Through comprehensive benchmarking testing using Hyperledger Caliper in a distributed multi-node environment, this research successfully answered the research questions and tested the proposed hypothesis.

Based on empirical data analysis obtained from 300+ test iterations, several main conclusions can be drawn:

1) **Trade-off Between Capacity and Responsiveness**: Results show contradictory performance characteristics between the two algorithms. **PoS excels in capacity aspect** (throughput) with peak achievement up to ≈60 TPS, almost twice as high compared to PoA which experiences saturation at ≈30–40 TPS. However, this advantage must be paid with lower responsiveness. **PoA proved far superior in latency**, recording average confirmation time of 4.6 seconds, significantly faster than PoS (≈8 seconds).
2) **Resource Efficiency**: In terms of operational efficiency, **PoA shows superiority** in computational resource utilization. PoA can produce transactions per second per CPU unit (TPS/%CPU) higher than PoS. This is caused by PoS's more complex architecture, involving two process layers (Execution Layer and Consensus Layer) and intensive gossip communication traffic between validators, thus creating additional computational overhead.
3) **Suitability with DChain Characteristics**: Considering academic certificate system characteristics that are Read-Heavy (verification dominance compared to issuance), User-Centric (requires fast response during verification), and has moderate transaction volume (seasonal issuance), **PoA consensus mechanism is deemed more relevant and effective** for implementation in DChain's early consortium stage. Although having lower maximum throughput, 30 TPS capacity is already very adequate to handle daily operational load of higher education institutions, while latency and efficiency advantages provide more crucial added value for user experience and operational costs.

Overall, this research concludes that although PoS offers promises of long-term scalability and wider decentralization, **PoA remains the most optimal technical solution** for DChain's current needs, balancing performance, cost efficiency, and response speed.

Future research should explore: (1) wider network variables with more nodes (> 20 nodes) and real WAN geographic conditions; (2) consensus parameter optimization such as reducing block time or increasing block gas limit; (3) security robustness analysis against cyber attack scenarios; and (4) hybrid model studies (PoA-PoS) where PoA serves as sidechain for fast daily operations with periodic checkpointing to main PoS network for long-term security.

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All source code, configurations, and datasets used in this research are publicly available at: https://github.com/Skripsi-Maulana-Anjari

REFERENCES

[1] W. Choi and J. W.-K. Hong, "Performance evaluation of ethereum private and consortium blockchain: Poa vs. pos," in *Proceedings of NOMS 2020 - 2020 IEEE/IFIP Network Operations and Management Symposium*. IEEE, 2020, pp. 1–6.

[2] G. A. F. Rebello, G. F. Camilo, L. C. B. Guimaraes, and F. L. Koch, "Performance evaluation of proof-of-authority and proof-of-stake consensus algorithms in private ethereum networks," in *Proceedings of the IEEE International Conference on Blockchain (Blockchain)*. IEEE, 2020, pp. 1–8.

[3] C. Arslan, S. Sipahioğlu, E. Şafak, and M. Gözütok, "Comparative analysis and modern applications of consensus algorithms in blockchain networks," *IEEE Access*, vol. 8, pp. 105 507–105 520, 2020.

[4] R. Li, "Comparative analysis and future directions of blockchain consensus algorithms: Pow, pos, bft, and dag," in *Proceedings of the 2021 International Conference on Computing, Communication, and Security (ICCCS)*. IEEE, 2021, pp. 35–39.

[5] D. K. K, D. N. Duraimutharasan, S. H. J. Shahana, and S. S. Kumar, "Performance evaluation of blockchain using hyperledger caliper," in *Proceedings of the 2020 6th International Conference on Advanced Computing and Communication Systems (ICACCS)*. IEEE, 2020, pp. 1–6.

[6] R. K. Kaushal and N. Kumar, "Exploring hyperledger caliper for performance benchmarking of blockchain," in *Proceedings of the 2020 International Conference on Computing for Sustainable Global Development (INDIACom)*. IEEE, 2020, pp. 1052–1056.

[7] R. Sharma, A. Bansal, and S. Bansal, "Performance analysis of private ethereum blockchain using consensus algorithms," *Procedia Computer Science*, vol. 173, pp. 341–350, 2020.

[8] S. Ali, M. I. Aslam, and M. Aslam, "Evaluation of blockchain consensus algorithms for performance and scalability," in *Proceedings of the 2020 International Conference on Computing, Networking and Communications (ICNC)*. IEEE, 2020, pp. 1–6.

[9] S. Gaikwad and K. Patel, "Blockchain based academic certificate verification system," in *Proceedings of the 2019 3rd International Conference on Computing Methodologies and Communication (ICCMC)*. IEEE, 2019, pp. 1216–1220.

[10] A. Singh and P. Chatterjee, "Securing academic records using blockchain technology," in *Proceedings of the 2019 IEEE International Conference on Electrical, Computer and Communication Technologies (ICECCT)*. IEEE, 2019, pp. 1–6.

[11] P. Wicaksono, P. Hatta, and Y. H. Aristyagama, "Comparative analysis of pos and poa consensus in ethereum environment for blockchain-based academic transcript systems," *Bulletin of Electrical Engineering and Informatics*, vol. 14, no. 3, pp. 2380–2392, June 2025.

[12] Hyperledger Foundation, "Hyperledger caliper documentation," https://hyperledger.github.io/caliper/, 2024, accessed: 2025-01-20.